

WORK PROCEDURE

PRO-OP-DD-012A

PROCEDURE FOR REMOTE DRILLING WITH A DIAMOND DRILL

OBJECTIVE

The objective of this procedure is to define the safe methods for remote drilling with a diamond drill.

APPLICATION

This procedure applies to all employees of MACHINES ROGER INTERNATIONAL (MRI) as well as to the client's geology department. Remote drilling should be decided by the client's geology department based on the hazards that may be present while drilling. The driller or MRI supervision are invited to propose remote drilling if they consider it safer than conventional drilling.

Article 440 of the Regulation respecting Occupational Health and Safety in Mines provides:

440. Notwithstanding section 439, holes may be drilled at distances less than those prescribed by that section provided that the drilling is carried out by a remote control device, under supervision, and that the drilling zone is evacuated.

NOTE: We apply the rules of Quebec unless the one that applies on the site is more severe

This procedure provides a framework for remote drilling work.

INTERPRETATION

The personnel having authority for the interpretation of this document is the director of operations. Any request for modification, revision or update must be made to this person.

RESPONSABILITIES

DRILLER

The driller must apply the remote drilling procedure according to the requirements proposed by the client. He must respect all the procedures surrounding drilling with a diamond drill or any other equipment. This procedure is the minimum requested by Machines Roger International.

PROCEDURE FOR REMOTE DRILLING WITH A DIAMOND DRILL

MRI SUPERVISOR

The MRI supervisor must ensure the application of this procedure as well as compliance with the customer's engineering plan and specifications. He must also participate in any risk analysis requested by MRI or by the client.

CUSTOMER'S GEOLOGY DEPARTMENT

The geology responsible for drilling must issue drilling plans and specifications remotely. These plans and specifications must take into consideration the risks associated with this type of drilling. The customer's engineering must be the initiator, if a risk analysis is necessary, or at least participate in this risk analysis.

MINE OPERATIONS

The mine operations department must require the application of this procedure. He must also participate in the implementation of work in collaboration with MRI and geology, in addition to participating in any risk analysis

PROCEDURE

1. Install the drill above the hole to be drilled according to the drilling plans and specifications
2. Install the system at a distance between the driller and the hole to be drilled at a minimum of 50' (15 meters):
 - a. The driller and the helper driller must be towards the exit, making sure that nothing is blocking access
 - b. Not be in the blast line, or
 - c. Some customers might also require to be behind protection such as:
 - i. A gallery perpendicular to the one where the drill is installed
 - ii. An equipment
 - iii. A pile of rock
3. All access to the drilling sites (driller and driller) must be barricaded by a fence or ea ribbon (according to customer requirements/procedures) or guarded by a worker
4. Signage indicating the type of work (e.g.: PROHIBITED ACCESS, DRILLING IN PROGRESS, REMOTE DRILLING.) must be posted on the fence or the ribbon.
5. This procedure must be put in a waterproof bag on the fences of all accesses and signed by the driller who will have visited these places before the start of drilling, at the start of each work shift.
6. If a visit to the drilling site is necessary, the operator must be contacted by radio, or if visual contact is possible, the driller will move to meet the person(s) wishing to visit the site.
7. The driller and the driller's helper will remain in the safe zone at all times while drilling.



Christian St-Amour
Operations Manager

PROCEDURE FOR REMOTE DRILLING WITH A DIAMOND DRILL

FEUILLE DE SIGNATURES DES BARRICADES LORS DU FORAGE À DISTANCE

Driller's signature :
Name (script letters) _____ Signature : _____

Date : _____ Night ☐ Day ☐ Time : _____ : _____

Driller's signature :
Name (script letters) _____ Signature : _____

Date : _____ Night ☐ Day ☐ Time : _____ : _____

Driller's signature :
Name (script letters) _____ Signature : _____

Date : _____ Night ☐ Day ☐ Time : _____ : _____

Driller's signature :
Name (script letters) _____ Signature : _____

Date : _____ Night ☐ Day ☐ Time : _____ : _____

Driller's signature :
Name (script letters) _____ Signature : _____

Date : _____ Night ☐ Day ☐ Time : _____ : _____

Driller's signature :
Name (script letters) _____ Signature : _____

Date : _____ Night ☐ Day ☐ Time : _____ : _____

Driller's signature :
Name (script letters) _____ Signature : _____

Date : _____ Night ☐ Day ☐ Time : _____ : _____

Driller's signature :
Name (script letters) _____ Signature : _____

Date : _____ Night ☐ Day ☐ Time : _____ : _____

Driller's signature :
Name (script letters) _____ Signature : _____

Date : _____ Night ☐ Day ☐ Time : _____ : _____